



UNIVERSITÀ
DI TORINO



Electron transport properties tutorial

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Transport properties in CRYSTAL

A semiclassical model is considered, the Boltzmann Transport Equation (BTE) equation is solved considering three main approximations:

- Fermi-Dirac occupation of bands
- Frozen band approximation
- Constant relaxation time approximation

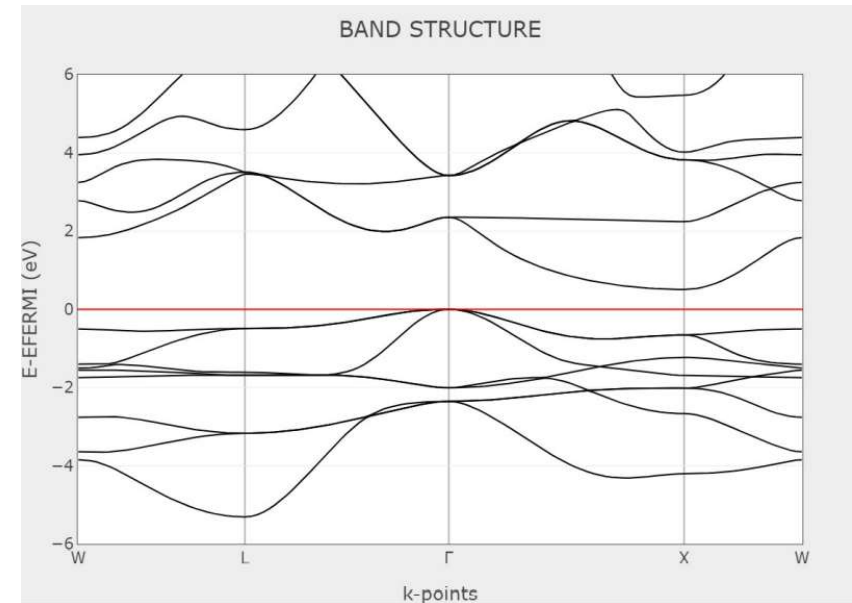
The aim of the model is to relate the band structure to the transport properties

Band structure $\epsilon_n(\mathbf{k})$

Band velocity $\mathbf{v}_{q,n}(\mathbf{k}) = \frac{\partial \epsilon_n(\mathbf{k})}{\partial k_q}$

Transport Distribution Function

$$\Xi_{qr} = \sum_n \sum_{\mathbf{k}} \tau_n[\epsilon_n(\mathbf{k})] \mathbf{v}_{q,n}(\mathbf{k}) \mathbf{v}_{r,n}(\mathbf{k})$$



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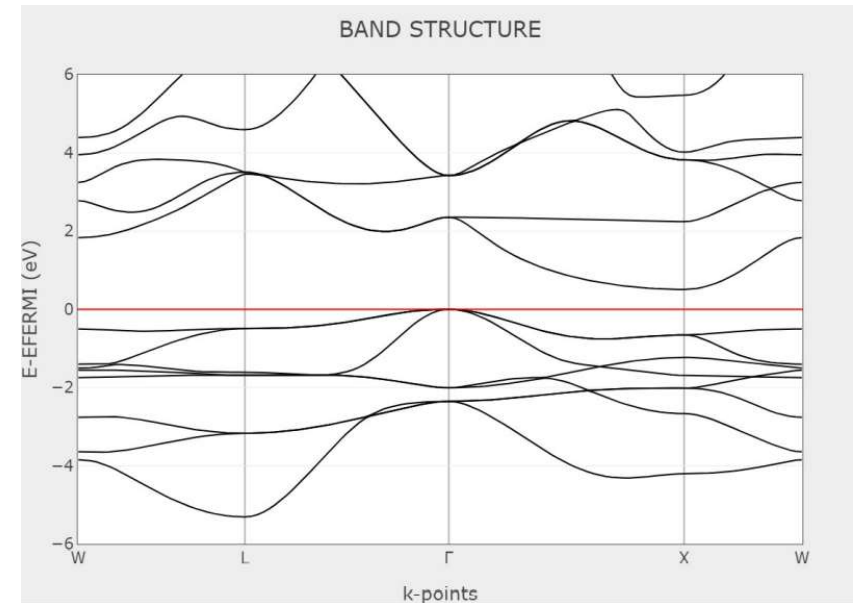
Transport Distribution Function

$$\Xi_{qr} = \sum_n \sum_{\mathbf{k}} \tau_n[\epsilon_n(\mathbf{k})] \mathbf{v}_{q,n}(\mathbf{k}) \mathbf{v}_{r,n}(\mathbf{k})$$

$$\Xi_{qr} = \tau \sum_n \sum_{\mathbf{k}} \mathbf{v}_{q,n}(\mathbf{k}) \mathbf{v}_{r,n}(\mathbf{k})$$

Constant relaxation time approximation

τ = relaxation time approximation, depend on scattering mechanism



Transport Coefficients

Solving the BTE in a semi-classical model:

$$\sigma(\mu, T) = \int \left(-\frac{\partial f}{\partial \epsilon} \right) \Xi(\epsilon) d\epsilon$$

Electron conductivity

$$\sigma \mathbf{S}(\mu, T) = \frac{1}{T} \int \left(-\frac{\partial f}{\partial \epsilon} \right) (\epsilon - \mu) \Xi(\epsilon) d\epsilon$$

Seebeck coefficient

$$k_e(\mu, T) = \frac{1}{T} \int \left(-\frac{\partial f}{\partial \epsilon} \right) (\epsilon - \mu)^2 \Xi(\epsilon) d\epsilon$$

Electron thermal conductivity

Experiment
Comparison!

μ = chemical potential, f = Fermi distribution function, ϵ = energy, Ξ = transport distribution function

General remarks

In order to obtain reasonable results for transport properties of materials few things must be kept in mind:

- A good SCF convergence is necessary;
- Dense **k**-sampling in reciprocal space are strongly recommended
- The Fermi level is computed and printed in the *properties* calculation; the centering of the plots is up to the user.

The input file

NEWK val val 1 0	Recalculation of the Hamiltonian eigenvectors Shrinking factors of the Monkhorst and Gilat net respectively Fermi energy calculation is performed, none additional printing
BOLTZTRA TRANGE minval maxval step MURANGE minval maxval step TDFRANGE minval maxval step [Optional sub-keywords] END	Keyword to calculate the electron transport properties Definition of the temperature range in Kelvin Definition of the chemical potential range in eV Definition of the transport distribution function (TDF) range in eV End of the BOLTZTRA input block
END	End of input properties calculation

Once the system wavefunction is computed with *crystal*, the electron conductivity, the Seebeck coefficient, and the electron thermal conductivity can be got by inserting the keyword **BOLTZTRA** after the **NEWK** input block. The eigenvectors are not stored at the end of *crystal* calculation thus the recalculation of them is compulsory. The **BOLTZTRA** option opens a sub-block that has to be closed with a keyword **END**.

The input file

NEWK val val 1 0	Recalculation of the Hamiltonian eigenvectors Shrinking factors of the Monkhorst and Gilat net respectively Fermi energy calculation is performed, none additional printing
BOLTZTRA TRANGE minval maxval step MURANGE minval maxval step TDFRANGE minval maxval step [Optional sub-keywords] END	Keyword to calculate the electron transport properties Definition of the temperature range in Kelvin Definition of the chemical potential range in eV Definition of the transport distribution function (TDF) range in eV End of the BOLTZTRA input block
END	End of input properties calculation

The **BOLTZTRA** option requires three mandatory keywords: **TRANGE**, **MURANGE**, and **TDFRANGE**. Optional keywords, as **RELAXTIM**, which permits to set a new value of relaxation time(default 10 fs), **SMEAR**, which sets a smearing factor for the transport distribution function, and **SMEARTYP**, which allows changing the kind of smearing method, can be added in the **BOLTZTRA** sub-block.

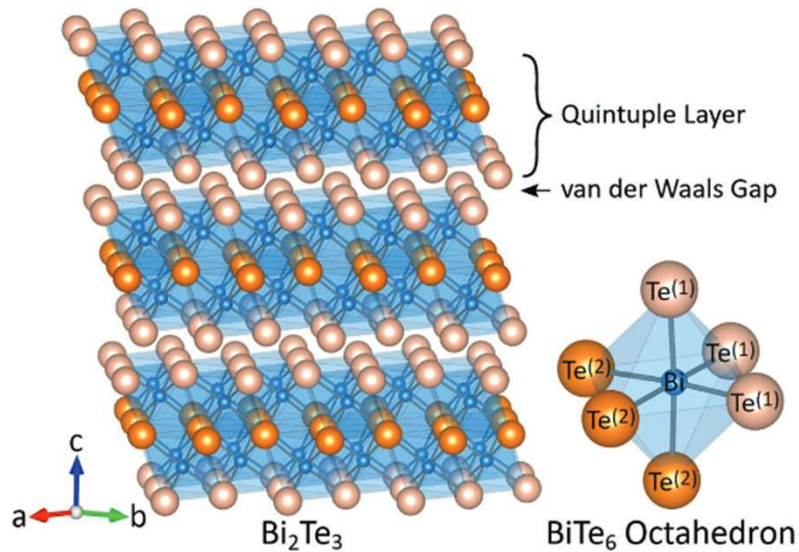
Choose the Shrinking Factor

$$\Xi_{qr} = \sum_n \sum_{\mathbf{k}} \tau_n[\epsilon_n(\mathbf{k})] \mathbf{v}_{q,n}(\mathbf{k}) \mathbf{v}_{r,n}(\mathbf{k})$$

As you can see from the definition, the transport distribution function contains a sum on $\mathbf{k}$ points: this is why a high shrinking factor is needed in calculating transport properties (higher than the shrink used in the SCF calculation). The optimal shrinking factor is system dependent.

If you want to choose the shrinking factor for your system you have to compute transport properties for increasing the shrinking factor until the properties are stable. The lower SH which allows you to have stable properties is your ideal factor.

Case study: Bi_2Te_3



Bi_2Te_3 is anisotropic

Bismuth telluride is the most important class of thermoelectric materials because they have the highest known efficiency near room temperature.

$$ZT = S^2 \sigma T / k$$

S = Seebeck coefficient
 σ = electrical conductivity
 T = temperature
 k = thermal conductivity
 $S^2 \sigma$ = power factor

Case study: Bi₂Te₃

Bi2Te3	Title
CRYSTAL	Dimensionality of the system
0 0 0	
166	Space Group
4.395 30.440	Cell parameter
3	Number of non equivalent atoms
83 0.000 0.000 0.4005	Atomic number and Cartesian coordinates of the symmetry-irreducible atoms
52 0.000 0.000 0.0000	
52 0.000 0.000 0.2097	
END	End of the geometry input block

```
DIIS ERROR: 0.23845E-07 AT SCF CYCLE 7 - DIIS ACTIVE
TTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTT FDIK TELAPSE 14.91 TCPU 13.75
INSULATING STATE
TOP OF VALENCE BANDS - BAND 14; K 1; EIG -1.5903025E-01 AU
BOTTOM OF VIRTUAL BANDS - BAND 15; K 1; EIG -1.4802264E-01 AU
DIRECT ENERGY BAND GAP: 0.2995 eV
TTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTTT PDIG TELAPSE 14.93 TCPU 13.77
CHARGE NORMALIZATION FACTOR 1.00000000
```

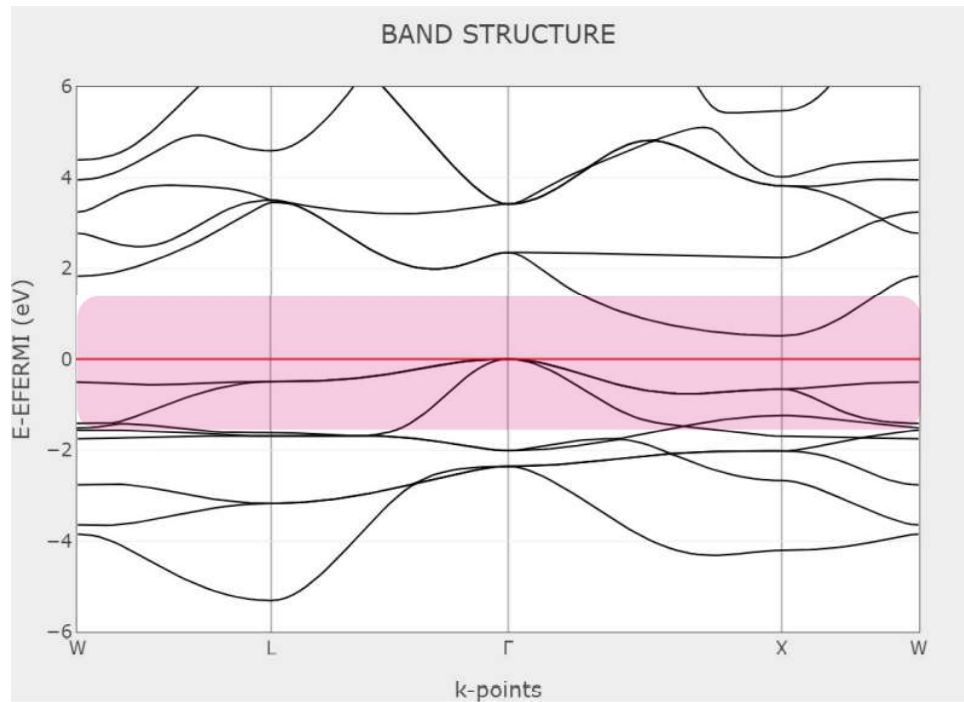
CRYSTAL INPUT

At first the wavefunction of the system has to be computed.

CRYSTAL OUTPUT

Check the SCF convergence and look for the energy of the top of the valence and bottom of the conduction band(- 4.5 and -4 eV).

Case study: Bi_2Te_3



Choose of chemical potential range:

Look for the top of the conduction band and the bottom of the conduction band. Then select a range 0.5 or 1 eV larger.

Case study: Bi₂Te₃

NEWK 48 48 1 0	Recalculation of the Hamiltonian eigenvectors Shrinking factors of the Monkhorst and Gilat net, respectively Fermi energy calculation is performed, none additional printing
BOLTZTRA TDFRANGE -5.5 -3 0.001 MURANGE -5.5 -3 0.01 TRANGE 300. 700. 200. RELAXTIM 22 END	Keyword to calculate the electron transport properties Setting of the TDF range in eV Setting of the chemical potential range in eV Setting of the temperature range in eV Setting of the relaxation time in fs End of the BOLTZTRA input block
END	End of the geometry input block

The ranges of the TDF and chemical potential can be to extend, at least, to 0.5/1 eV more the energies of the top valence and bottom virtual levels (which were -4.5 and -4 eV). About the steps of these ranges, higher values for the TDF mean a better integration calculation (much higher computational cost) while higher values for the chemical potential mean a better "graphical resolution". Relaxation time has to be taken from literature.

Output files

From a BOLTZRA calculation we obtain different files:

- .outp contains informations about the calculation
- SEEBECK.DAT stores the Seebeck coefficient data
- SIGMA.DAT stores the electron conductivity data
- KAPPA.DAT stores the electron thermal conductivity data

Case study: Bi_2Te_3

From a BOLTZRA calculation we obtain different files:

- .outp
- SEEBECK.DAT
- SIGMA.DAT
- KAPPA.DAT

Let's look at them together!

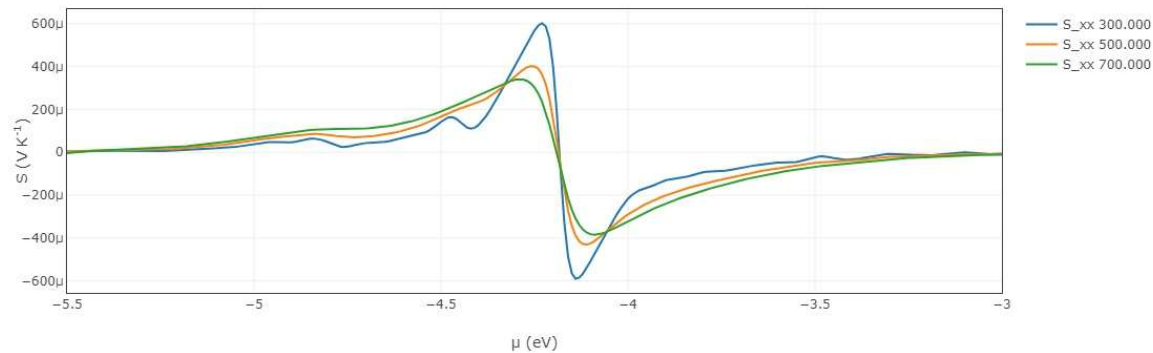
Exercise: try to visualize these three properties with **CRYSLOT** or with **CRYSTALClear**!

Visualizing results

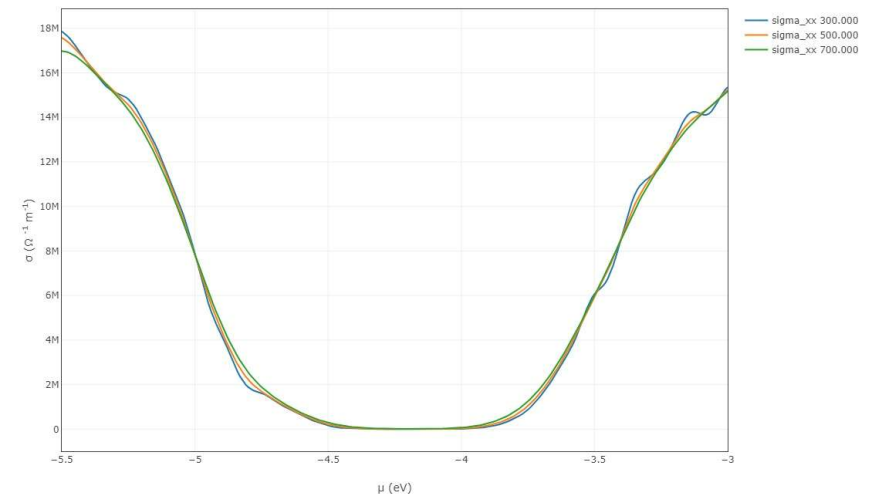
With **CRYSLOT** <https://crysplot.crystalsolutions.eu/index.html>



SEEBECK COEFFICIENT

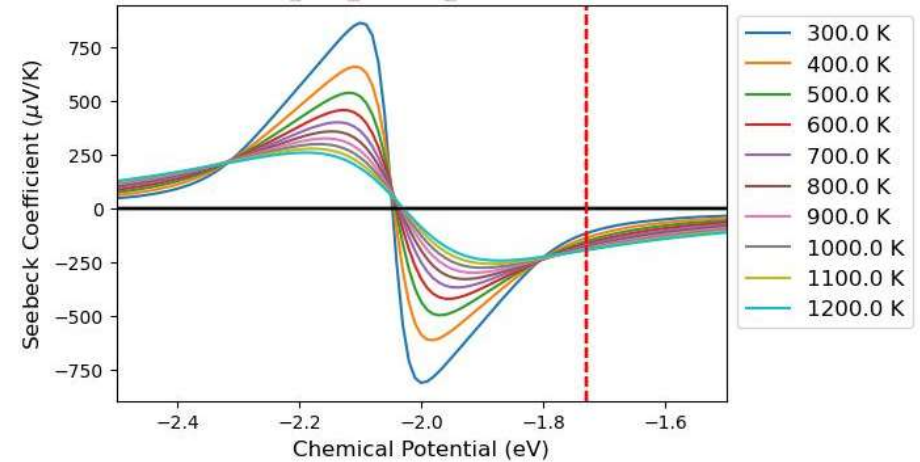
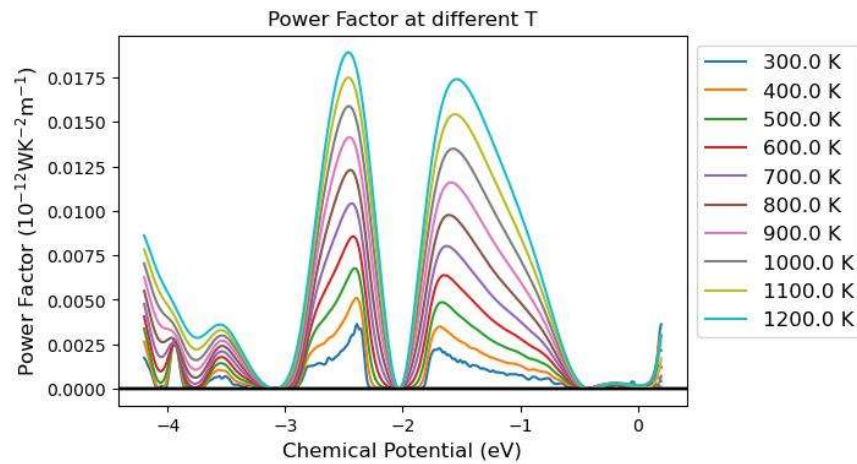


ELECTRON CONDUCTIVITY



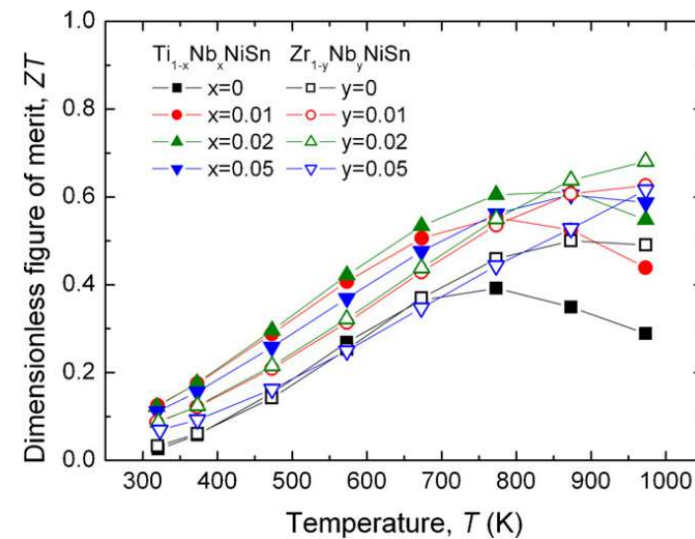
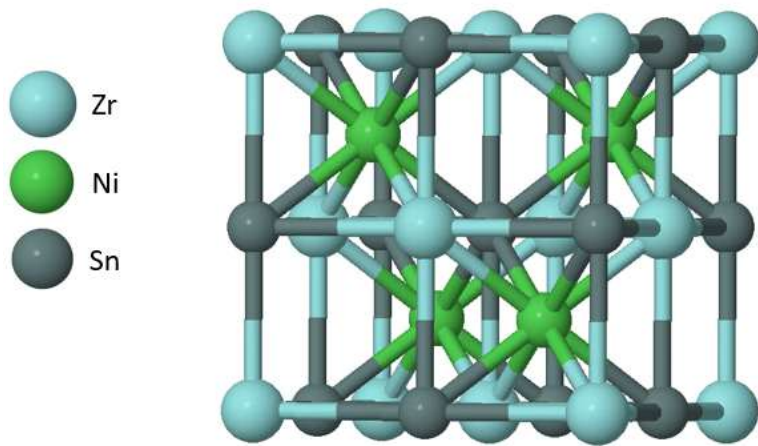
Visualizing results

With **CRYSTALClear** <https://github.com/crystaldevs/CRYSTALClear>



Case study: ZrNiSn

Imagine you want to study the thermoelectric Half Heusler alloy ZrNiSn.

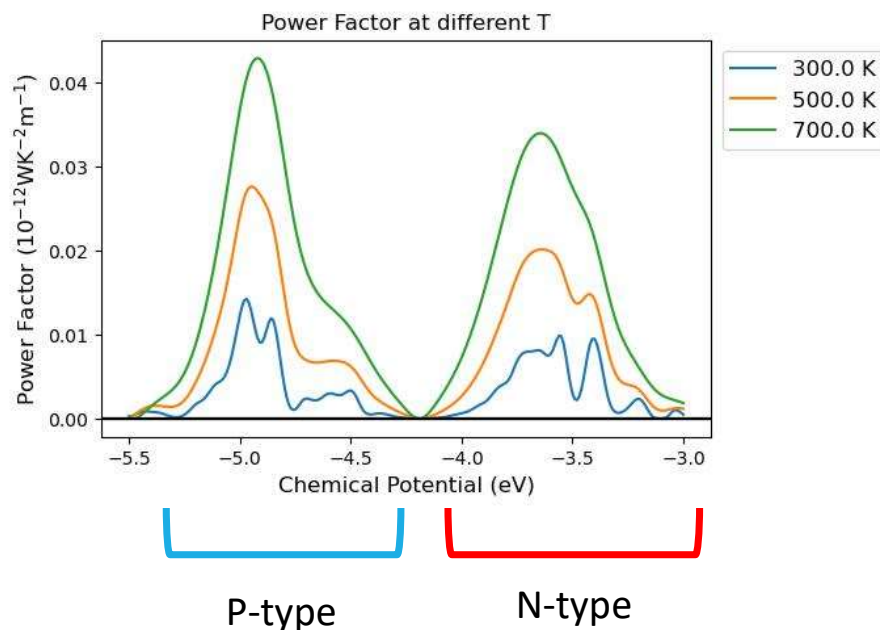


Work Strategy

1. Is the material an n-type or a p-type conductor?
2. Search in literature for transport properties data to set computational parameters: the relaxation time
3. Search in literature for transport properties data to compare with calculations
4. What is the transport properties temperature behaviour?
5. Do defects matter?

Work Strategy, step 1: is the material an n-type or a p-type conductor?

You can look at the power factor:



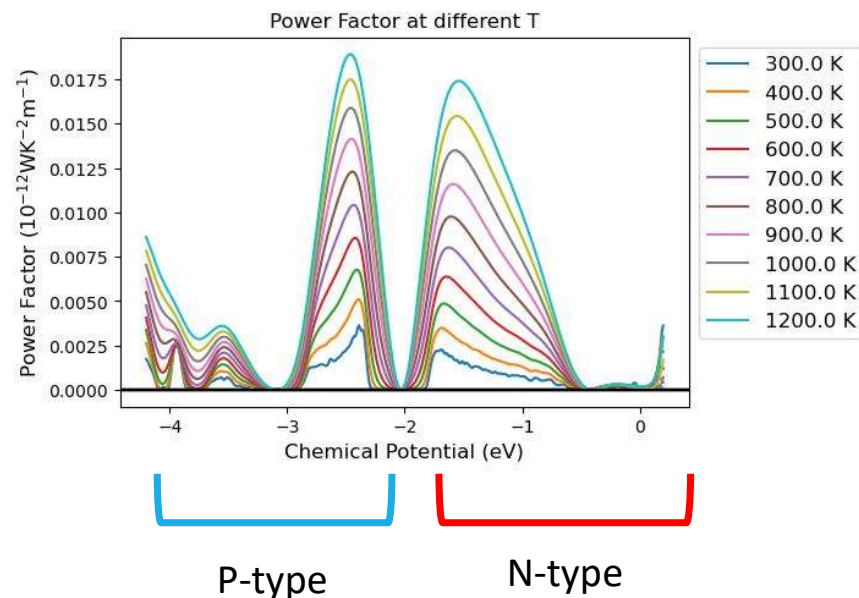
If the PF peak is higher in the p-type region the material is a p-type conductor, otherwise it is a n-type.

Exercise: with the .DAT i gave you tell me if ZrNiSn is p-type or n-type conductor.

Work Strategy, step 1: is the material an n-type or a p-type conductor?

You have to look at the power factor:

Theory



Experiment

Can the system actually be both a p- or an n- type conductor?

Experimentally, ZrNiSn is reported to be an n-type.

Work Strategy, step 2: set computational parameters

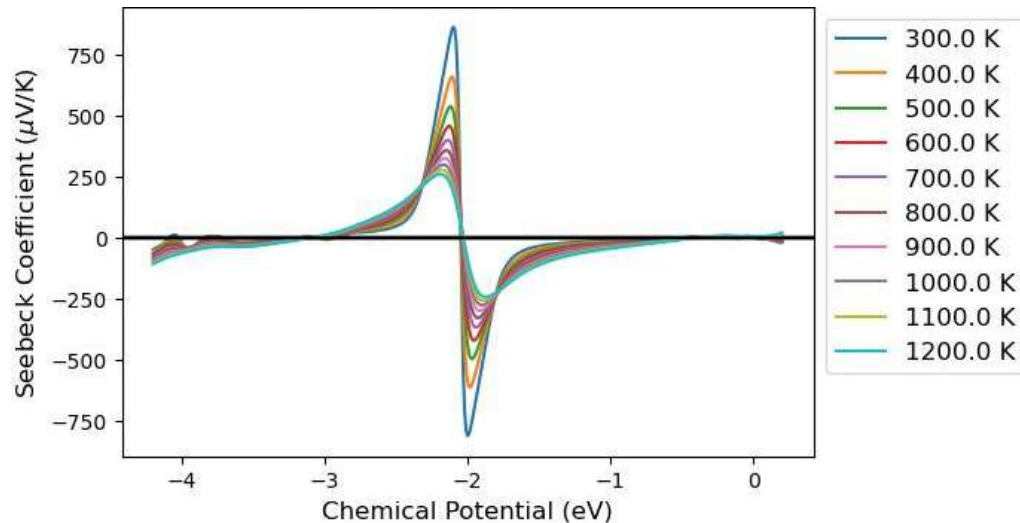
Imagine you want to study the thermoelectric Half Heusler alloy ZrNiSn. To set computational parameters (i.e. the relaxation time) you should compare yourself to literature.

```
NEWK
60 60
1 0
BOLTZTRA
TDFRANGE
-4.2 0.2 0.01
MURANGE
-4.2 0.2 0.01
TRANGE
100. 1200. 100
RELAXTIM
18
END
END
```

I provided you input and output files. Now try it yourself to visualise for example the computed transport properties!
How could you know if obtained results are correct?

E. Ascrizzi et al.: Materials 17, 1061 (2024)

Work Strategy, step 3: compare with literature



From CRYSTAL we easily obtain Seebeck coefficient data versus the chemical potential. Experimental studies on the other hand may report only absolute values of S , his temperature behaviour and hopefully the charge carrier concentration of the sample.

For ZrNiSn for example in literature study the charge carrier concentration is about 5 or $4 \times 10^{19} \text{ cm}^{-3}$ and it is known to be an n-type conductor.

E. Ascrizzi et al.: Materials 17, 1061 (2024)

Work Strategy, step 3: compare with literature

To validate your results you should compare yourself to literature. Task: how do you compare your ZrNiSn Seebeck coefficient with the experimental one reported in figure?

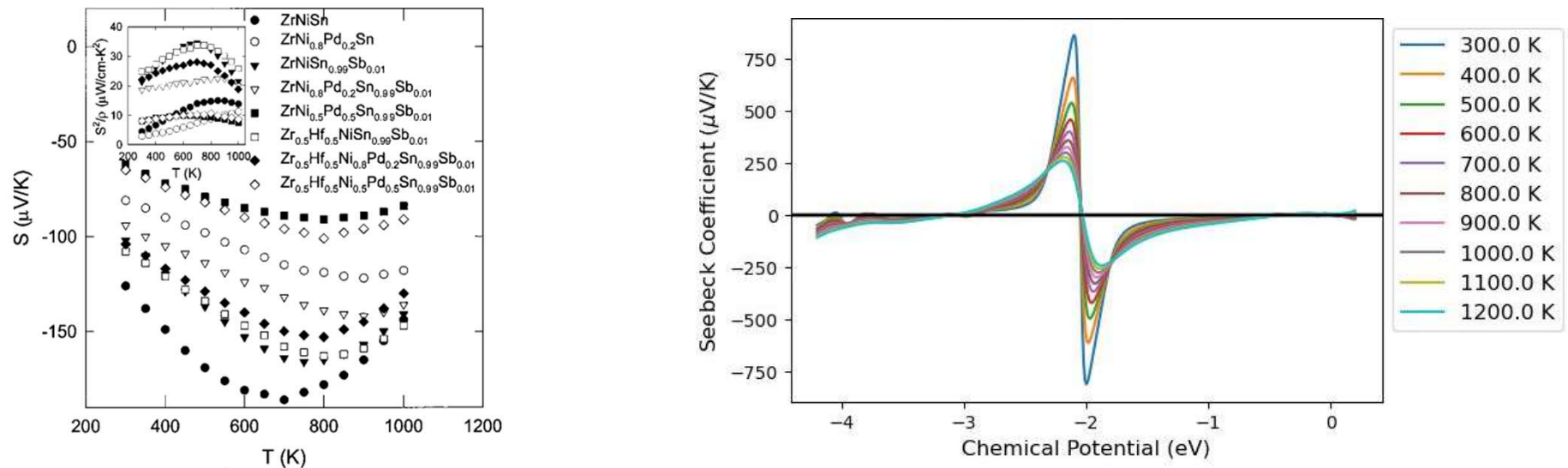


FIG. 2. Temperature dependence of the Seebeck coefficient and the power factor (insert) of ZrNiSn-based half-Heusler compounds.

E. Ascrizzi et al.: Materials 17, 1061 (2024)

Q. Shen et al., Appl. Phys. Lett., 79(25):41654167, Dec 2001.

Work Strategy, step 3: compare with literature

So, the experimental data we take as reference is:

$$S = -125 \mu\text{V/K at } CC \text{ } 5 \text{ or } 4 \times 10^{19} \text{ cm}^{-3}$$

Let's look together to a .DAT file and look for this information

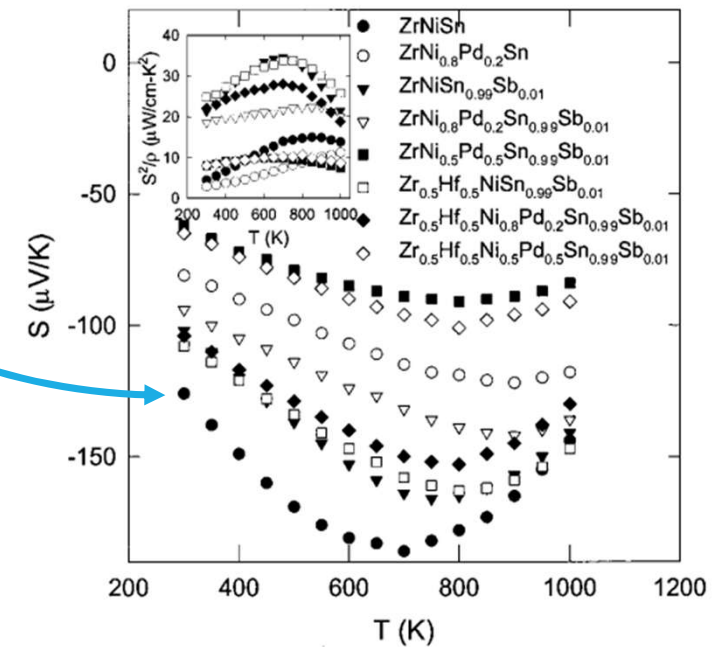


FIG. 2. Temperature dependence of the Seebeck coefficient and the power factor (insert) of ZrNiSn-based half-Heusler compounds.

Work Strategy, step 3: compare with literature

Columns in .DAT: Chemical potential, Temperature, Number of Carrier, Sxx, Sxy, Sxz...

```
# 0 [Seebeck coefficient in SI units, i.e. in V/K]
# Mu(eV) T(K) N(#carriers) S_xx S_xy S_xz S_yx S_yy S_yz S_zx S_zy S_zz
# T(K) = 300.00 Npoints = 441 V(cm^3) = 3.848E-22
```

-4.20	300.000	9.856E+00	-4.8307684E-05	-2.3378080E-18	-2.3535990E-18
-4.19	300.000	9.833E+00	-4.4941995E-05	-2.8328396E-18	-2.8477330E-18
-4.18	300.000	9.805E+00	-4.1849083E-05	-3.3752019E-18	-3.3876034E-18
-4.17	300.000	9.773E+00	-3.8899441E-05	-3.9330432E-18	-3.9424016E-18
-4.16	300.000	9.734E+00	-3.5907869E-05	-4.4615418E-18	-4.4683158E-18
-4.15	300.000	9.688E+00	-3.2663908E-05	-4.9188093E-18	-4.9240236E-18
-4.14	300.000	9.634E+00	-2.8981503E-05	-5.2833364E-18	-5.2879881E-18
-2.04	300.000	-1.200E-07	-2.5815901E-04	-5.8952732E-17	-5.8787367E-17
-2.03	300.000	-3.725E-07	-5.4248586E-04	-4.1676712E-17	-4.1679669E-17
-2.02	300.000	-6.814E-07	-7.1185968E-04	-2.4492515E-17	-2.4557643E-17
-2.01	300.000	-1.094E-06	-7.8955575E-04	-1.3046348E-17	-1.3120865E-17
-2.00	300.000	-1.671E-06	-8.1199818E-04	-6.7537812E-18	-6.8248842E-18
-1.99	300.000	-2.502E-06	-8.0547148E-04	-3.6013343E-18	-3.6686462E-18
-1.98	300.000	-3.711E-06	-7.8475343E-04	-2.0904065E-18	-2.1553941E-18
-1.97	300.000	-5.482E-06	-7.5730641E-04	-1.3814942E-18	-1.4452239E-18

E. Ascrizzi et al.: Materials 17, 1061 (2024)

Work Strategy, step 3: compare with literature

Columns in .DAT: Chemical potential, Temperature, Number of Carrier, Sxx, Sxy, Sxz...

```
# 0 [Seebeck coefficient in SI units, i.e. in V/K]
# Mu(eV) T(K) N(#carriers) S_xx S_xy S_xz S_vx S_vy S_vz S_zx S_zy S_zz
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-4.20	300.000	9.856E+00	-4.8307684E-05	-2.3378080E-18	-2.3535990E-18
-4.19	300.000	9.833E+00	-4.4941995E-05	-2.8328396E-18	-2.8477330E-18
-4.18	300.000	9.805E+00	-4.1849083E-05	-3.3752019E-18	-3.3876034E-18
-4.17	300.000	9.773E+00	-3.8899441E-05	-3.9330432E-18	-3.9424016E-18
-4.16	300.000	9.734E+00	-3.5907869E-05	-4.4615418E-18	-4.4683158E-18
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-2.01	300.000	-1.094E-06	-7.8955575E-04	-1.3046348E-17	-1.3120865E-17
-2.00	300.000	-1.671E-06	-8.1199818E-04	-6.7537812E-18	-6.8248842E-18
-1.99	300.000	-2.502E-06	-8.0547148E-04	-3.6013343E-18	-3.6686462E-18
-1.98	300.000	-3.711E-06	-7.8475343E-04	-2.0904065E-18	-2.1553941E-18
-1.97	300.000	-5.482E-06	-7.5730641E-04	-1.3814942E-18	-1.4452239E-18

The experimental data we can compare with is the Charge Carrier Concentration, defined as:

Number of Carrier / Volume

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Work Strategy, step 3: compare with literature

If we just look at the column **Ncarrier** we can see that some of them are positive and some of them are negative: this is a convention. Being ZrNiSn a n-type conductor, look for a CC of -5 or $-4 \cdot 10^{19} \text{ cm}^{-3}$.
Exercise : which chemical potential at 300 K correspond to this carrier concentration?

```
# 0 [Seebeck coefficient in SI units, i.e. in V/K]
# Mu(eV) T(K) N(#carriers) S_xx S_xy S_xz S_yx S_yy S_yz S_zx S_zy S_zz
# T(K) = 300.00 Npoints = 441 V(cm^3) = 3.848E-22
```

-4.20	300.000	9.856E+00	-4.8307684E-05	-2.3378080E-18	-2.3535990E-18
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-2.04	300.000	-1.200E-07	-2.5815901E-04	-5.8952732E-17	-5.8787367E-17
-2.03	300.000	-3.725E-07	-5.4248586E-04	-4.1676712E-17	-4.1679669E-17
-2.02	300.000	-6.814E-07	-7.1185968E-04	-2.4492515E-17	-2.4557643E-17
-2.01	300.000	-1.094E-06	-7.8955575E-04	-1.3046348E-17	-1.3120865E-17
-2.00	300.000	-1.671E-06	-8.1199818E-04	-6.7537812E-18	-6.8248842E-18
-1.99	300.000	-2.502E-06	-8.0547148E-04	-3.6013343E-18	-3.6686462E-18
-1.98	300.000	-3.711E-06	-7.8475343E-04	-2.0904065E-18	-2.1553941E-18
-1.97	300.000	-5.482E-06	-7.5730641E-04	-1.3814942E-18	-1.4452239E-18

P-type: Ncarrier positive

N-type: Ncarrier negative

E. Ascrizzi et al.: Materials 17, 1061 (2024)

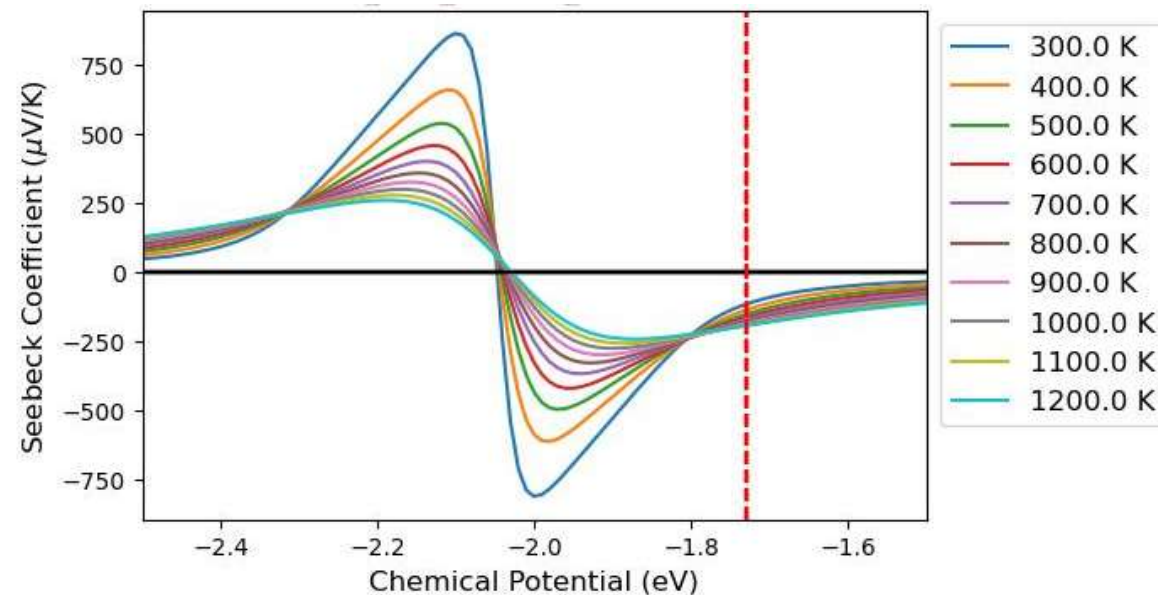
Work Strategy, step 4: check the temperature behaviour

What I found:

A carrier concentration similar to what experimentally reported: $-3.6 \cdot 10^{19} \text{ cm}^{-3}$ corresponding to $\mu = -1.73 \text{ eV}$.

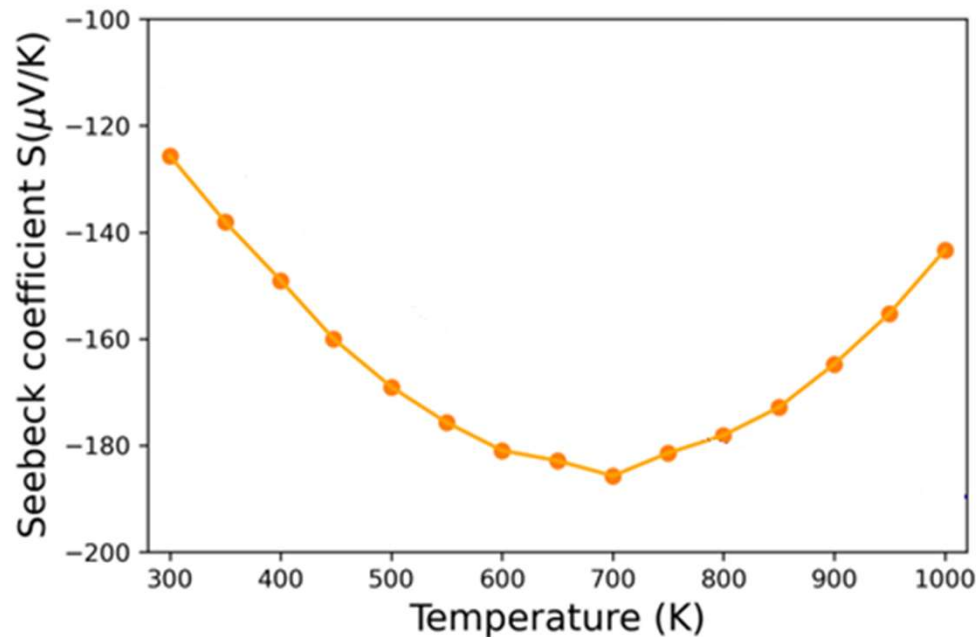
Here $S = -114 \mu\text{V/K}$ (exp. From Shen et al. $-125 \mu\text{V/K}$).

Now we have to check the temperature dependence. Exercise: go to see the Seebeck coefficient value at $\mu = -1.73 \text{ eV}$ at different temperatures.



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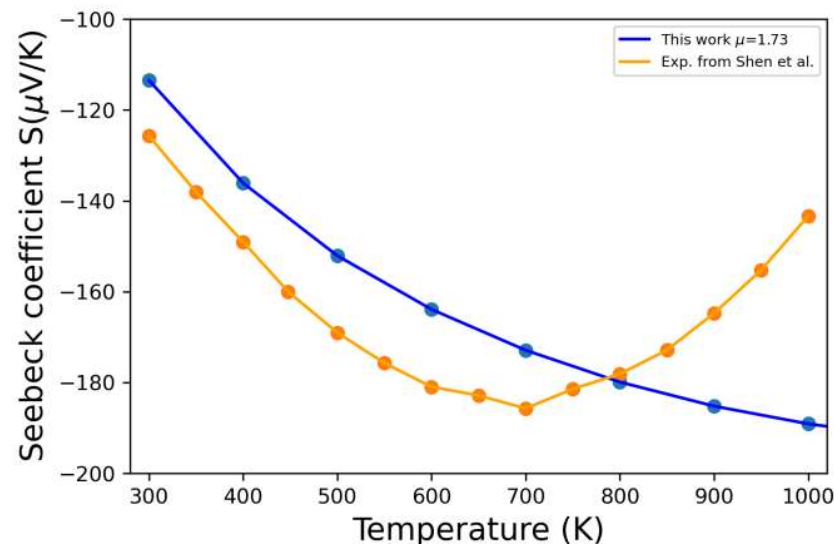
Work Strategy, step 4: check the temperature behaviour



This is the Seebeck temperature behaviour reported by Shen et al. Are your data similar to these ones? Make a plot with your Seebeck coefficients at different temperatures at $\mu = -1.73$ eV.

Work Strategy, step 4: check the temperature behaviour

I plotted Seebeck values at different temperatures at $\mu = -1.73$ eV together with data from Shen et al. We can see that there is agreement in the first part of the graph



For Shen et al. decrease after a maximum is reached seems to be due to the excitation of electron-hole pairs across the energy gap, and the opposing contribution to S from the two carriers reduces the observed S . We cannot see it with CRYSTAL because excitation is not considered.

E. Ascrizzi et al.: Materials 17, 1061 (2024)

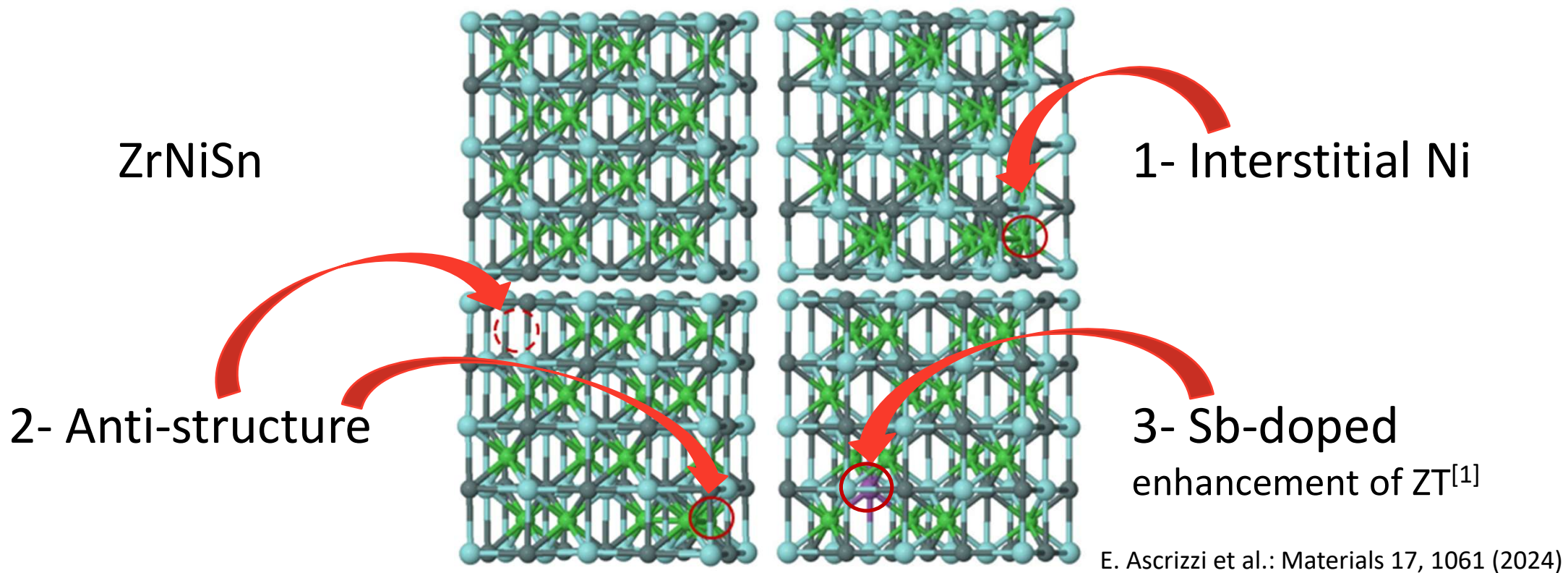
Work Strategy, step 5: defects and doping

Another important thing to consider comparing theoretical data to experimental ones is that experimental samples will never be perfect.

Intrinsic defects should be considered as they modify transport properties. Furthermore if doping is simulated: suggestions to improve thermoelectric properties can be made.

One should examine literature in order to find which defects and in which concentrations are worth studying.

Work Strategy, step 5: defects and doping



^[1] H. Xie, H. Wang, C. Fu, Y. Liu, G. J. Snyder, X. Zhao, and T. Zhu. The intrinsic disorder related alloy scattering in ZrNiSn half-Heusler thermoelectric materials. Sci. Rep., 4(6888):16, Nov 2014.

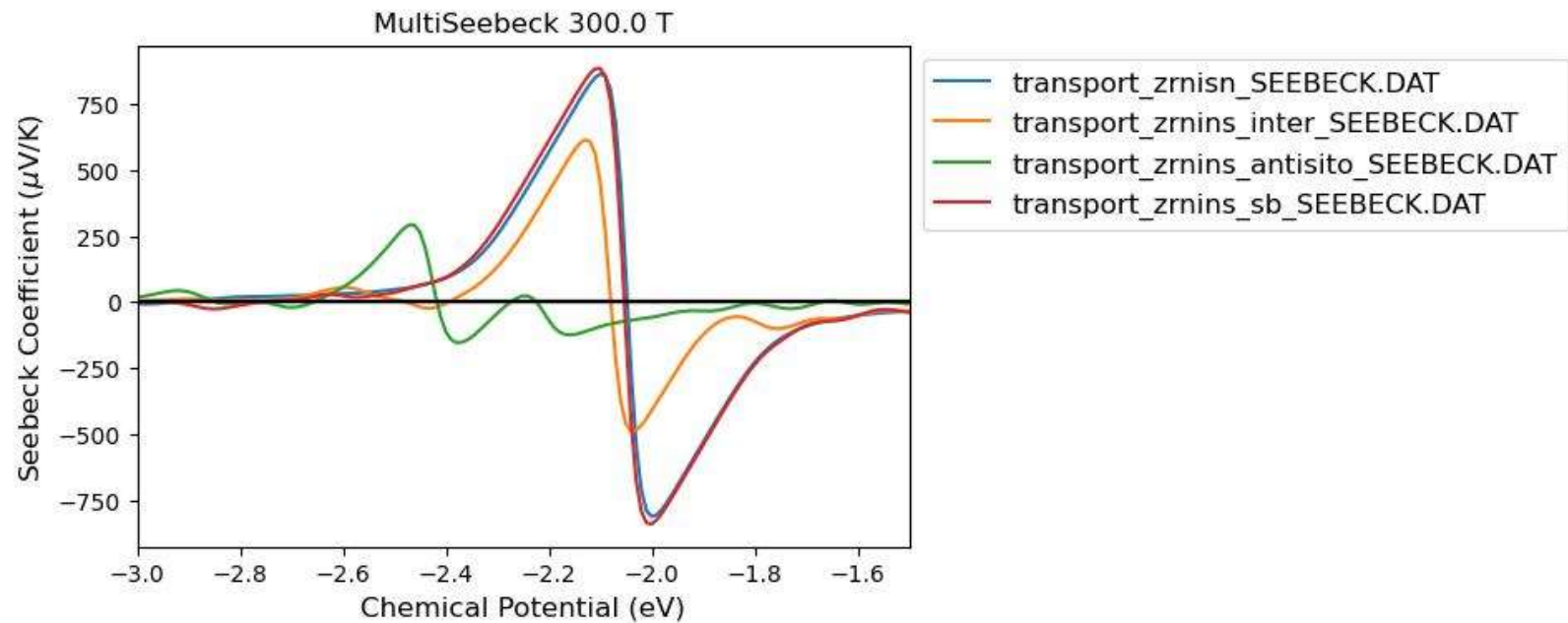
Work Strategy, step 5: defects and doping

I provided you the .DAT files for the perfect structure of ZrNiSn, ZrNiSn with an interstitial Ni atom, with an antisite Ni atom and doped with Sb. Now, try it yourself to determine how defects influence the electron transport properties of this alloy. To do so, you can read the .DAT files, use **CRYSLOT** or **CRYSTALClear**.

If there's still time: fill up a table comparing S , σ , κ and PF at 300 K, $\mu = -1.73$ eV.

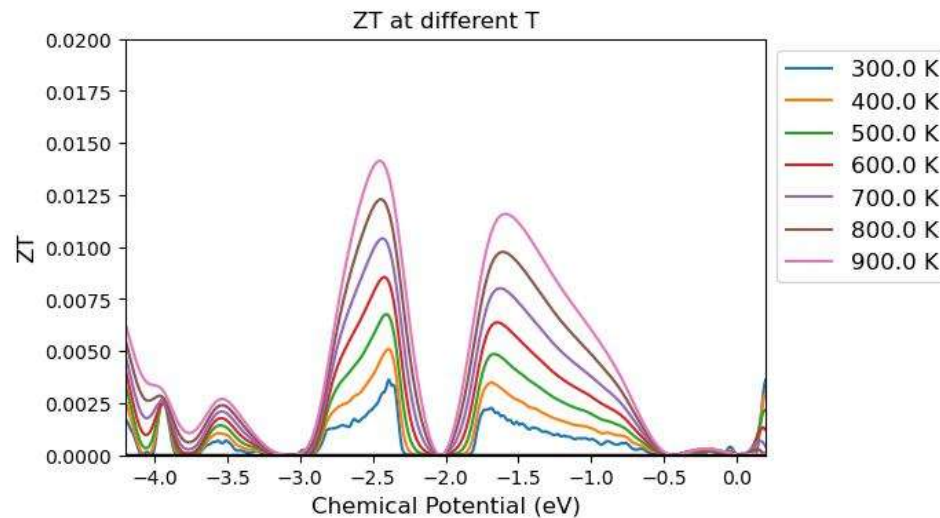
System	S [$\mu\text{V/K}$]	σ [S/m]	κ_e [$\text{Wm}^{-1}\text{K}^{-1}$]	PF [$10^{-12}\text{WK}^{-2}\text{m}^{-1}$]
ZrNiSn				
Interstitial Ni				
Antisite Ni				
Sb doping				

Work Strategy, step 5: defects and doping

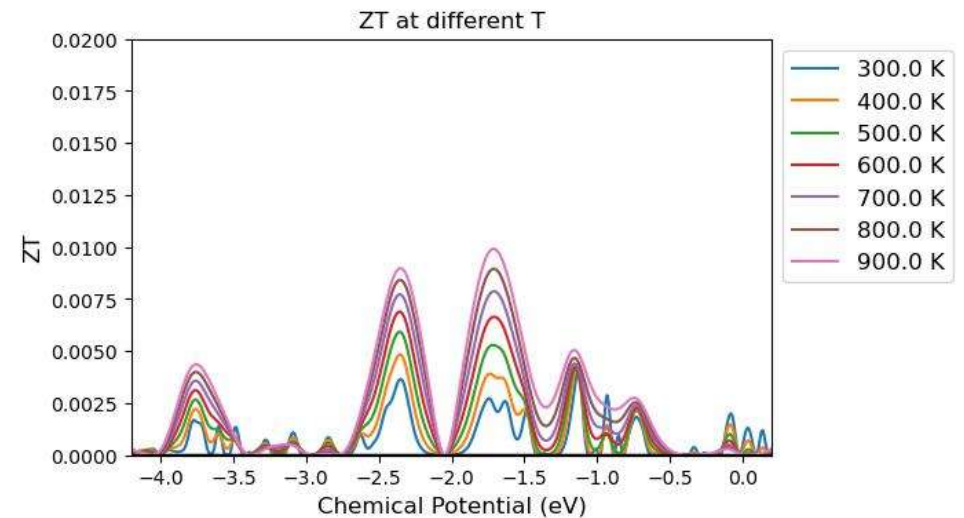


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Work Strategy, step 5: defects and doping



ZrNiSn $k_{\text{tot}} = 6 \text{ WK}^{-1}\text{m}^{-1}$



ZrNiSn doped with Sb $k_{\text{tot}} = 9 \text{ WK}^{-1}\text{m}^{-1}$

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How to modify and contribute to CRYSTALClear

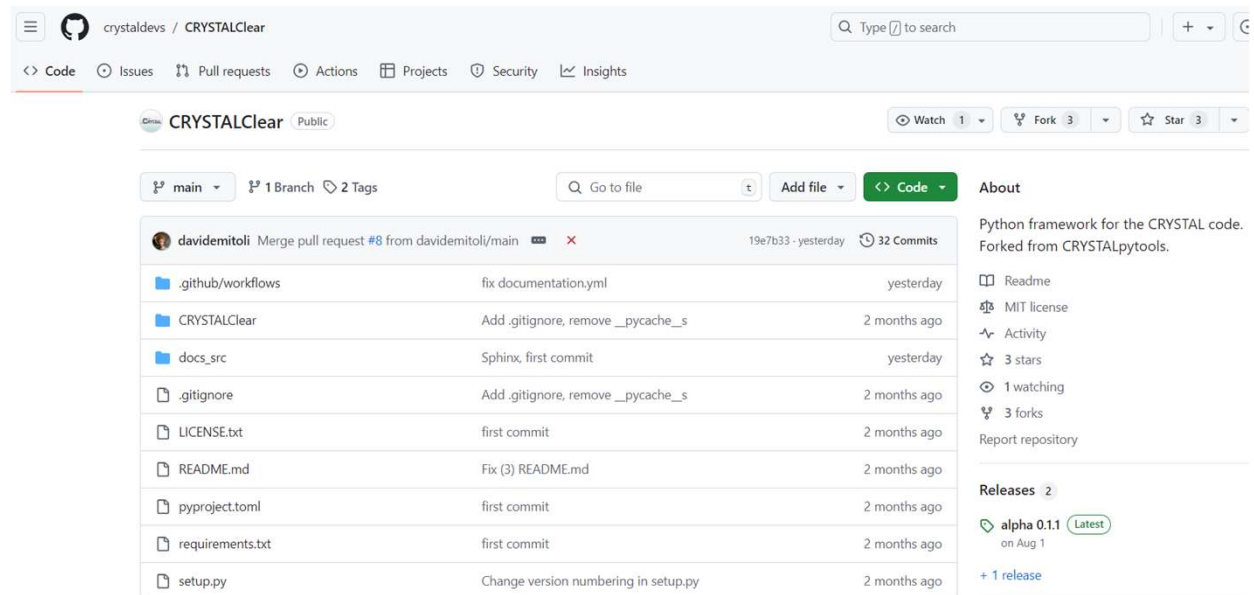
The crystal_functions are on Github

<https://github.com/crystaldevs/CRYSTALClear>

Clone the repository and modify as you prefer:

As a user: customize your plots

As a developer: you can open a pull request



Thank you for your attention!